

A Lock Free, Non-blocking, In Line Processing Architecture for High Throughput and Regulatory Compliant Blockchain Trading Applications

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andrew.p.legear@ul.ie | ECSA 2026, Bolzano, Italy | Artifact: Zenodo DOI 10.5281/zenodo.21512313

Motivation

- Regulated decentralised finance (DeFi) requires trading systems that are simultaneously **high-throughput** and **regulatory compliant**.
- Off-chain compliance checks (KYC/AML, trade surveillance) and on-chain settlement pull system design in opposite directions.
- Existing blockchain trading architectures are not empirically benchmarked against production-grade throughput and latency targets.^[1]
- Research question:** which architectural pattern maximises throughput while preserving compliance guarantees, under real blockchain constraints (nonce ordering, gas limits, block capacity)?

Approach

- Two-layer **hybrid architecture**: off-chain compliance / order layer decoupled from on-chain settlement layer.
- Ablation study** of five progressively optimised architectural variants, deployed and measured on a live production trading system.
- Variants isolate the effect of: asynchronous submission, multi-threading, multi-wallet parallelism, and symbol-level sharding with lock-free coordination.
- Empirical measurement of throughput, latency (avg / P99), and block utilisation for each variant under identical load.

Two-Layer Hybrid

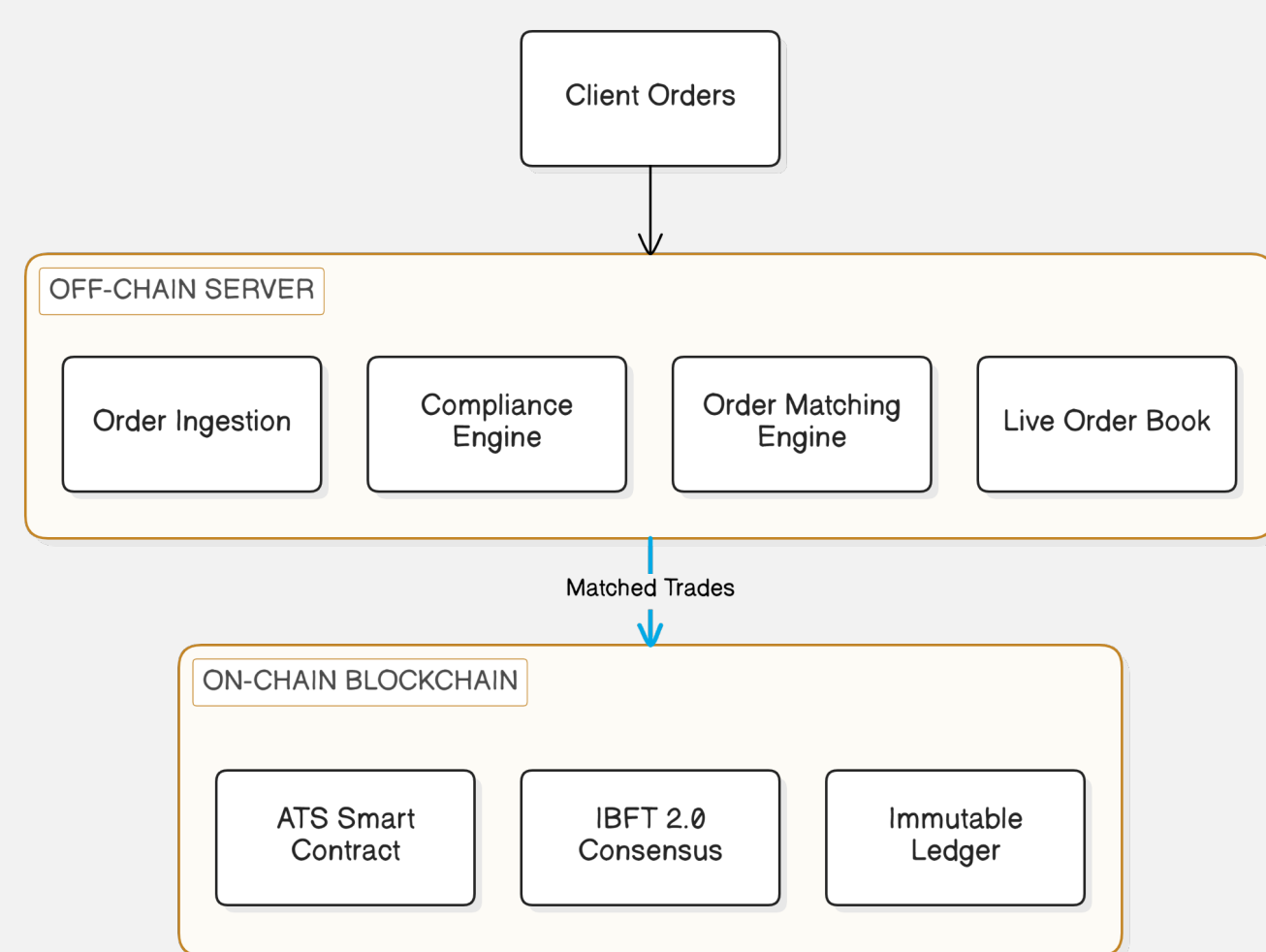


Fig. 1: 2-layer hybrid architecture.

Off-chain compliance, matching, and order-book components are decoupled from on-chain settlement, keeping regulatory logic mutable while the ledger stays immutable.

Five architectural stages studied:

- Sequential, blocking, single wallet
- Asynchronous, single wallet
- Multi-threaded, asynchronous, single wallet
- Unsynchronised multi-threaded, multi-wallet
- Symbol-sharded, lock-free, multi-wallet (proposed)**

Industry Profile

- Horizon Globex Ireland DAC** is a fintech services company delivering blockchain-backed trading and exchange systems.
- Deployments include US automated trading systems and the **Upstream** platform in the Seychelles.
- Upstream supports **MERJ**, a tier-one regulated stock exchange.

Key Findings

- Nonce mechanism** limits naive parallelism to only a **36% throughput gain** — Ethereum's strict per-account nonce ordering is a hard serialisation point.^[4]
- Circumventing nonce ordering with multiple wallets **without** careful coordination causes a **9× increase in latency** due to lock contention.
- Bottlenecks **migrate**: solving one on-chain constraint exposes a new server-side concurrency crisis.^[2]
- Our **lock-free, symbol-sharded** submission model (inspired by HFT architectures) removes contention entirely by partitioning work per trading symbol.^[3,6]

Performance Results

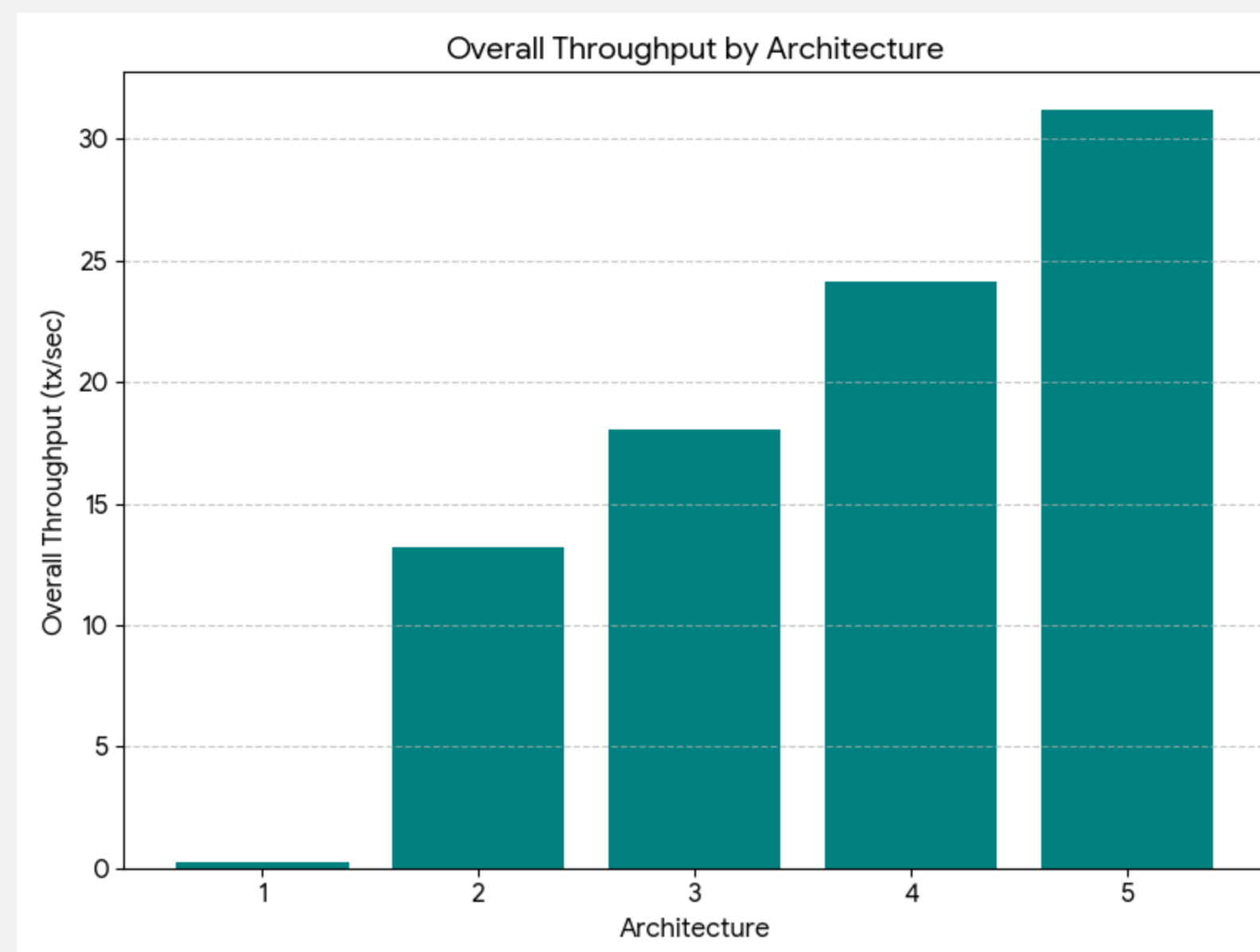


Fig. 2: Throughput across architectural stages.

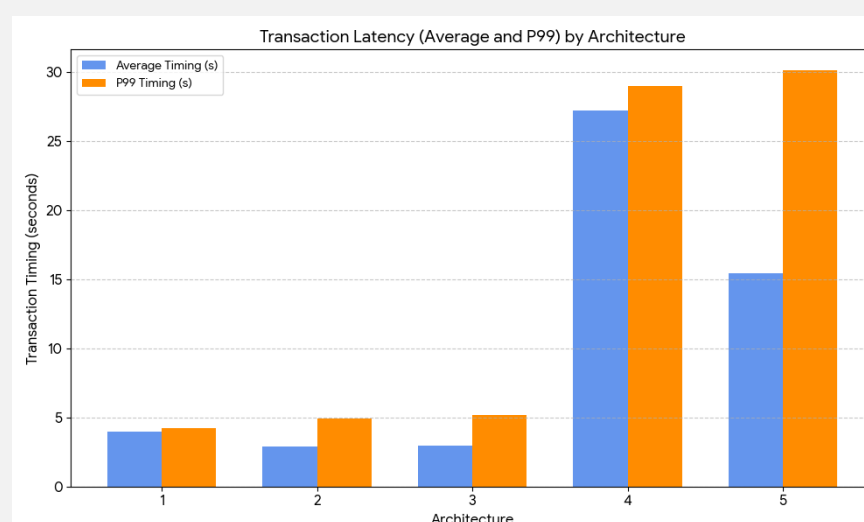


Fig. 3: Latency

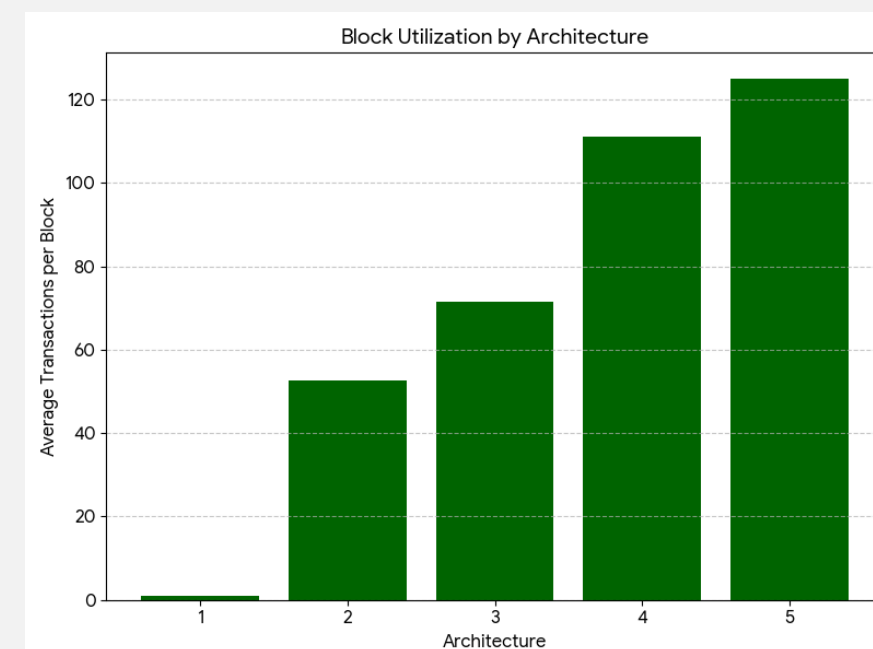


Fig. 4: Block utilisation

2.8× improvement over conventional design
125× gain over naive baseline
44% latency reduction at peak throughput

Why It Matters

- Demonstrates that **holistic co-design** of on-chain and off-chain components is essential — optimising either layer in isolation is insufficient.
- Provides an empirically validated architecture pattern directly applicable to production regulated-DeFi trading systems.
- Highlights blockchain-specific constraints (nonce ordering, gas limits, block capacity) as first-class architectural drivers, not implementation details.

Production System

- Upstream** is Horizon Globex's live platform, running **Architecture 5** (symbol-sharded, lock-free) in production.
- Off-chain server handles order ingestion, KYC/AML, and matching.
- Settlement via permissioned **IBFT 2.0** Ethereum and the **ATS.sol** contract.

Symbol-Sharded, Lock-Free Design

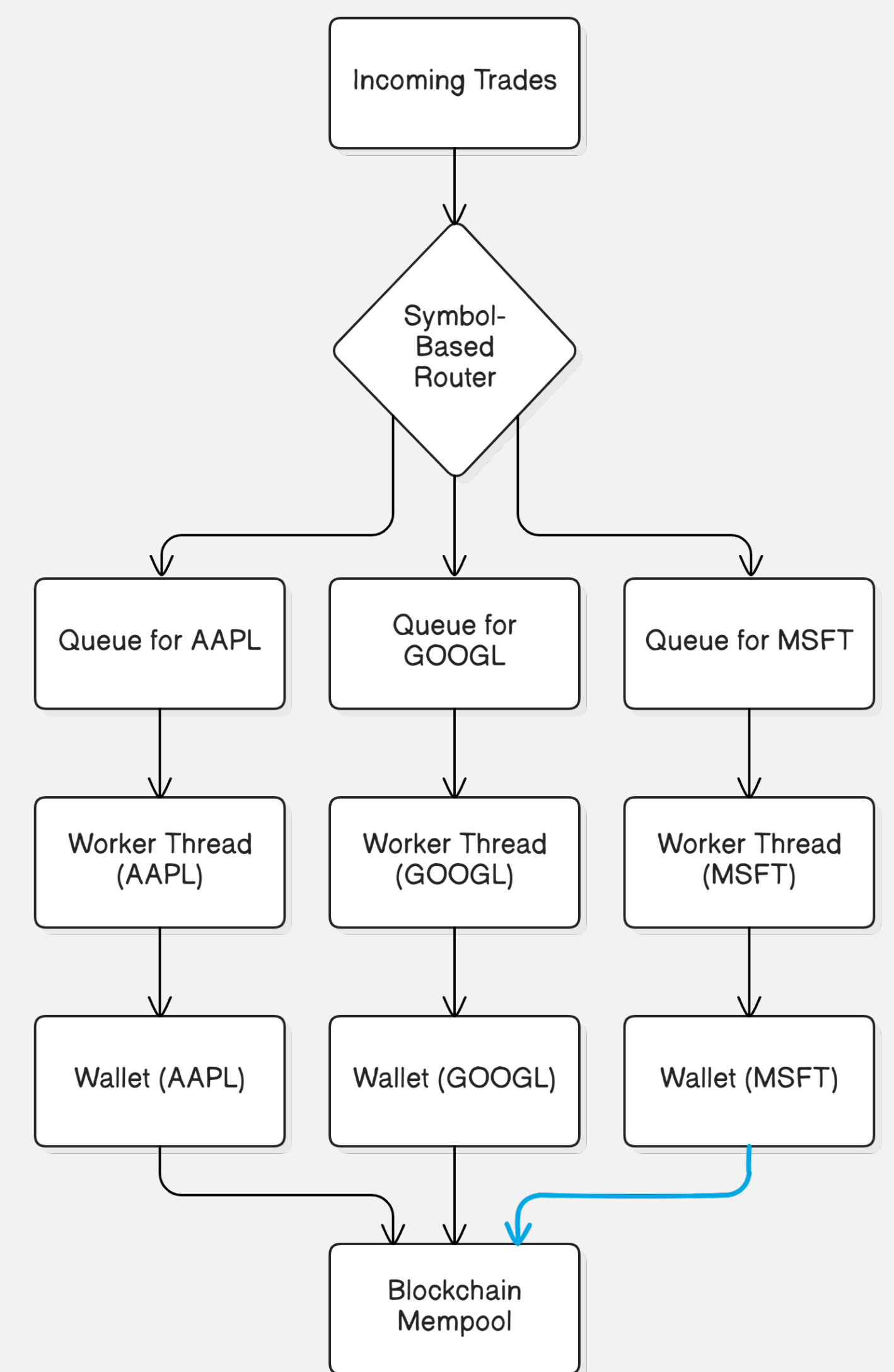


Fig. 5: Symbol-sharded, lock-free design.

Trades are routed by symbol to independent queues and worker threads, each owning a dedicated wallet, removing cross-symbol lock contention entirely.

Conclusions

- A symbol-sharded, lock-free architecture achieves **31.19 tx/sec**, the highest throughput across all five studied variants.
- Naive concurrency fixes (multi-threading, multi-wallet) expose new bottlenecks rather than resolving throughput ceilings.^[2,4]
- Regulatory-compliant, high-throughput blockchain trading is achievable through careful **architectural co-design**, not brute-force parallelism.
- Future work: adaptive sharding and cross-chain generalisation.

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